Responsible Tailings Management

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PURPOSE OF THE DOCUMENT

This position statement sets out HZL's approach to the governance of tailings storage facilities (TSFs) to minimise the risk of catastrophic failure of tailings facilities. Tailings are a common by-product of the mining process which are generated as waste during the process of separating out the valuable mineral from mined ore. Responsible tailings management constitutes one of the material risks of mining industry, owing to its multidisciplinary and complex nature. In case of failures of any kind, the consequences would be catastrophic to human life as well as the environment.

With climate change resulting in increasing number of extreme weather conditions and changes in average rainfall, the risk of tailings facility failure is more if not constructed, operated, and maintained to meet the best global practices. In this context, the Global Industry Standard on Tailings Management (GISTM) jointly released by International Council on Mining and Metals (ICMM), United Nations Environment Program (UNEP) and Principles for Responsible Investment (PRI) in August 2020, is a significant development in the international mining sector. This document provides guidance on managing tailings storage facilities (TSFs) throughout its life including design, operation, closure, and post-closure.

At HZL, we are committed to manage our TSFs in a safe and environmentally responsible way, without compromising on the health and well-being of our employees as well as communities. We regularly undertake consultations with local communities and other stakeholders of interest to incorporate and address their concerns regarding tailings management

IMPACT

A tailings incident has the potential to have a devastating impact on communities and the surrounding environment. Tailing management plans are an essential prerequisite for sound storage practice, as most failures of tailing storages around the world result from inadequate management of the storages. Effective implementation of a management plan not only results in a safer tailing storage facility (TSF) but will frequently reduce overall costs associated with operation and closure of the facility. The main focus of management of tailings facilities is on ensuring failures do not occur. HZL currently manages 3 tailings facilities and all 3 are in active use.

HZL has set up Dry Tailing Plant which is based on separating water from tailings slurry generated in the beneficiation process. Company repurposes tailings materials and waste rock as backfill to stabilise our underground mining operations, while the remaining tailings are then placed in a specially designed tailings storage to minimise the environmental, social, and economic risks. Key benefits of the dry tailing technology include recirculation of more than 80% of the process water present in tailings, a faster rehabilitation and restoration of storage site at mine closure and ensuring re-availability of water for further use. Hindustan Zinc has implemented global leading practices in tailing dam management, and we are associating with renowned global experts to provide long-term monitoring and advice on the design, construction, and operation of our three tailing storage facilities. Overall good construction quality, along with upgradation of tailing slurry transport and discharge lines through installation of piezometers (and inclinometer) with automated data collection and web-based monitoring, make it world-class.

GOVERNANCE

There is Tailing storage facility committee at corporate level and at all the three mining sites which is comprising of representatives of various functions. By strengthening our governance model and clarifying the chain of accountability, HZL has TSF Manager who is responsible for:

- Creating Awareness on tailing management policy, standard, plan and best practices
- Ensuring adherence of Operations, Maintenance and Surveillance (OMS) Manual
- Ensuring effectiveness of reclaim water system
- Ensuring effectiveness of Emergency Planning and Response
- Ensure the availability of all required resources
- Ensure Continuous monitoring and reporting

POLICY & STANDARD

HZL has a dedicated **Tailing Management Policy** that provides guidance on managing tailings and storage facilities to ensure zero harm to human life and environment. We also have an organisation wide **Tailing Management Standard** that provides a framework on the approach, methodology and guidance on tailings management to be followed in our operations. All our operations are in compliance with the National and International laws/Regulations/Guidelines as well as our own **Environment Policy**.

In order to ensure there are no deviations from our overall approach, we believe that governance is required at all levels of organization – site, corporate, and external parties. Our governance framework has been developed based on the ICMM guidelines as well as **Vedanta's Tailing Management Facility (TMF)** Standard. HZL ensures compliance with Vedanta's TMF Standard, which includes guidelines on following key aspects:

We aim to be fully compliant with the Global Industry Standards on Tailings Management (GISTM) by FY 2024-25

Six Key Elements of TSF Governance Framework as per ICMM

1. Accountability, Responsibility & Competency
2. Planning and Resourcing
3. Risk Management
4. Change Management
5. Emergency Preparedness and Response
6. Review and Assurance

TMF Planning and Design:

- Site selection with considerations on Environmental and social impact assessment findings, economics, as well as public health and safety.
- Dam break analysis to quantify TMF-related risks
- TMF design based on best available technology as per ICMM and other international guidelines
- TMF Construction with supervision and quality assurance provided by third-party consultant

TMF Operation and Maintenance:

- Tailings Management Committee with defined roles and responsibilities
- Tailings Management Plan
- Periodic risk assessments and mitigation plans
- Emergency Response Plan
- TMF audits according to performance standard

All our operating mines have a Tailing Management Plan – an Operations, Maintenance and Surveillance (OMS) manual.

Our facilities are regulated, permitted, and have been managed for many years to comply with local laws, regulations, permits, licenses, and other requirements. Our internal assurance processes verify that our managed TSFs operate in accordance with the Vedanta standard and also in alignment with Global Industry Standard on Tailing Management. Our operational TSFs have emergency response plans – tested through training exercises in collaboration with stakeholders. They follow strict business resilience and communication protocols

STRATEGY

HZL's overall governance approach to tailings aims to maintain the capacity of our facilities, ensure the qualifications of the people responsible for ensuring safety, and ensure design & stability of structure. We use these three framing concepts to evaluate personnel, projects, operations, and initiatives and programs. Tailings storage facilities at all our operations meet or exceed regulatory requirements, including permit and licence conditions and other relevant or appropriate requirements. Our strategy on responsible tailings management broadly includes following aspects from both technical and governance perspective:

- Tailing Management Policy & Tailing Management Plan
- Tailing Storage Design
- Incorporating Best Practices
- Surveillance of Tailing Dams
- Internal / External Review
- Emergency Planning and Response

HZL'S ACTIVE TSFS



Rampura Agucha Mine



Rajpura Dariba Mine



Zawar Mine

TAILING MANAGEMENT PLAN

Tailing Management Plan - Operations, Maintenance and Surveillance (OMS) Manual, that has been developed for all the three mines describes the following aspects:

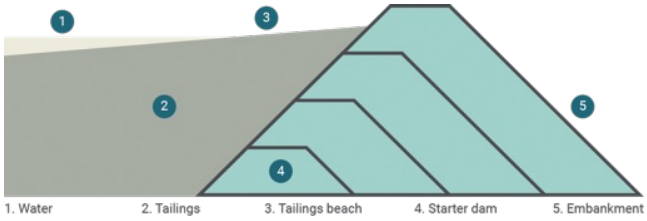
- Roles and responsibilities of TSF Committee
- Engineering and design aspects of the tailing facilities
- Procedures and processes for operation & maintenance, construction, surveillance of the facility; monitoring the components, their parameters
- Emergency procedures

Effective implementation of this management plan will not only result in a safer tailings storage facility (TSF) but will frequently bring down the overall costs associated with operation and closure of the facility.

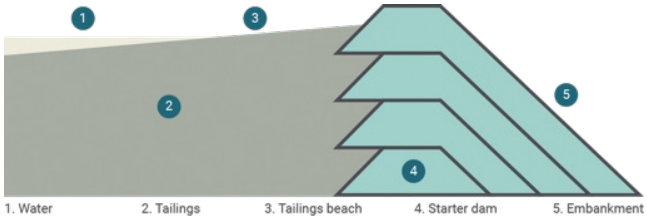
TAILING STORAGE DESIGN

We recognize the uniqueness of our mining locations and accordingly consider factors such as local topography and climate, the geotechnical, geological, and seismic activity, its proximity to people and infrastructure, land rehabilitation, and future land use while designing our storage facilities.

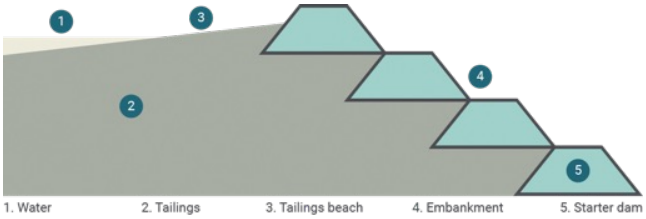
Downstream design continues the construction of the embankment wall on the external side of the compound with the batter extending on to what we call natural surface (or land that previously has not been disturbed), by depositing more support rock and bunding against the previously built tailings wall, thereby expanding the footprint, width and height of the initial embankment wall.



Center line design involves the continuation of the embankment construction on the external side of the compound however, does rely on the deposited tailings adjacent to the top of the initial embankment wall to support the raises.



Upstream design allows the embankment wall to be constructed on the inner side of the compound and relies on the strength of the deposited tailings to support the structural material for each lift of the embankment. This method allows for smaller quantities of earth and rock fill to be built on top of the tailings and therefore does not expand the footprint of the facility.



Based on the above said factors, we choose the best method from among the three main tailings construction methods, i.e., downstream, centreline, and upstream design.

TECHNOLOGY INTERVENTION

HZL is committed to integrating new and innovative technologies as part of our overall tailings management process. We make use of established and emerging dewatering approaches for applicable locations, as well as enhanced surveillance to add to our already extensive monitoring and response program across all of our facilities.

INCORPORATING BEST PRACTICES

The following are few best practices developed and incorporated at our three TSFs with an objective of continuously improving our tailings management:

- Clear policy on tailings management, and strong management commitment for safe and responsible
- Management of Tailings Storage Facility (TSF)
- TSF Committee – in-house experts to strengthen compliance to Vedanta TMF standard
- Reputable engineering and design firms selected for the design of these facilities
- Periodic TSF risk assessment conducted, and mitigation plan developed to minimize associated risks
- Tailings utilization in back filling through Paste fill/ Hydro fill
- Replacement of wet tailings disposal system with dry tailings disposal
- Supernatant water collected and recycled in process
- Garland Drains around tailings dam, Zero Liquid Discharge maintained from tailings dam
- HZL introduced a novel, satellite-based Interferometric Synthetic Aperture Radar (InSAR) monitoring technique to provide early warning of surface ground movements.



Supernatant Water Collection Pond



Garland Drains Around Tailing Dam



Dry Tailing plant

SURVEILLANCE OF TAILING DAMS

Surveillance involves inspection and monitoring of the operation, structural integrity, and safety of a facility. It consists of both qualitative and quantitative comparison of actual to expected behaviour. Regular review of surveillance information can provide an early indication of performance trends that, although within specification, warrant further evaluation or action.

1. Visual Monitoring and Inspection

- Weekly/Monthly and annual monitoring of the impoundment, dam safety instrumentation monitoring, and environmental monitoring by TSF committee

2. Geotech Monitoring on Tailing Dam

- Pillar-prism-total station data for measurement of Slope displacement
- Piezometer for measurement of pore water pressure
- Inclinometer for measurement of Sub-surface deformation
- Cameras are placed for security purpose monitoring

INTERNAL/EXTERNAL REVIEW

Internal Review	External Review
• Comprehensive internal audits/ inspection by cross-functional teams • Hazard identification & risk assessment conducted and reviewed periodically	• Engagement of world class consultants Indian Institute of Science (IISc) Bangalore, IIT Guwahati, etc. • Independent assessment by global experts Golder Associates to review the integrity/ stability of our dam structures and their associated management practices

EMERGENCY PLANNING & RESPONSE

Emergency Preparedness and Response Plan is available at all the three mines and is enabling us to identify emergency and hazardous conditions threatening the TSF, expedite effective response actions to prevent failure, and reduce loss of life and property damage. These systems, which demonstrate leadership and commitment to responsible tailings management practices, were developed through consultation with communities. We continually review our facilities and procedures and are committed to maintaining the highest standard of safety at our operations. See Our Performance Standard on **Emergency Planning and Preparedness** for further information.

Key features of Emergency Planning and Response are as follows:

- Identify emergency and hazardous conditions threatening the TSF
- Assessing, mapping, and mitigating the potential impacts from a hypothetical failure scenario
- Training and awareness to operations personnel in responding to emergency conditions
- Testing of effectiveness through validation drills
- Process of reporting on TSFs' failures

METRICS & PERFORMANCE

Goals

Complete transition from wet tailing to dry tailing disposal by 2025

Performance

- Zero incidents associated with existing tailings facilities
- 100% of active tailings facilities have undergone independent third-party reviews in the last three years
- No issues indicating potential risk identified in site tailings internal/ external review
- Recovery of water from tailing dam over FY 2022-23 was at 13 million m^3.
- Tailing utilisation was 5.55 Million Tons in FY 2022-23 due to reuse in backfilling and consistent operation.

High Risk Potential Tailing Sites

	Total Number of Facilities	Number of facilities categorized as "high risk potential" ("extreme" and "very high")	Percentage of "high risk potential" sites
Active Facilities	3	3	100%
Inactive Facilities (including facilities in care maintenance or closed)	1	0	
Planned Facilities	0	0	

TAILING DAM INVENTORY

S NO.	DESCRIPTION	RAMPURA AGUCHA	DARIBA	ZAWAR	DEBARI
1	Tailing facility Name	RA Mines Tailing dam	HZL - RDM complex RDC Tailing dam	Zawar Tailing storage facility	Jarosite Pond III
2	Location (Latitude / longitude coordinates)	25°50'53.5"N 74°44'25.7"E	24°57'58.7"N 74°08'51.0"E	24°20'32.34" N 73°42'45.58" E	24°35'40.9"N 73°49'24.1"E
3	Ownerships (Owned and operated, subsidiary, JV)	Operated 65% owned	Operated 65% owned	Operated 65% owned	Operated 65% owned
4	Status (Active, Inactive)	Active	Active	Active	Inactive
5	Date of initial operation	1991	1982	1982	2001
6	Is the dam currently operated or closed as per approved dam design	Yes	Yes	Yes. Dry stack method in process	Disposal Stopped Since Jul 2019.Yet to be closed as per approved design
7	Raising method (Upstream, centre line, modified centre line, downstream, landform etc.)	Phase 1 to Phase 7 – Fully Downstream, Phase 8 – Complete downstream except in a streah of 600 m raised with upstream Phase 9-combination of downstream & centreline method	Phase 1 to Phase 9 – Downstream and partially U/S , Phase 10 is under execution with D/S Method	Phase 1 to 5 with Downstream and phase 6 to 8 with Upstream	Upstream & Downstream
8	Current maximum height (m)	60 m	32 m	60 m through wet disposal mode followed by 40m dry stacking (434 mRL to 474 mRL	17 m
9	Current Tailing Storage impoundment volume (Million m^3)	57.8 million m^3 (Filled) & 5.9 Million m3 free space available	20 Million m^3	27.96 Million m^3	No storage allowed; capping is planned
10	Planned Tailing Storage impoundment volume (Million m^3) in 5 years (January 2024)	3.6	8	13.1	0
11	Most recent independent expert review	Dec 2022	Aug 2021	April 2022	Sept 2021
12	Do you have full and complete relevant engineering records including design, construction, operation, maintenance, and/ or closure	Yes	Yes	Yes	Yes
13	What is your hazard categorization of this facility, based on the consequence of failure	Category A	Category A	Extreme	Yet to be classified
14	What guideline do you follow for the classification system	ICOLD	ICOLD	Canadian Dam Association	HW rules 2016

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15	Has this facility, at any point in its history, failed to be confirmed or certified as stable, or experienced notable stability concerns, as identified by an independent engineer (even if later certified as stable by the same or different firm)	No	No	No	No
16	Do you have internal / in house engineering specialist oversight of this facility? Or you have external engineering support for this purpose?	External	External	External	External
17	Has formal analysis of the downstream impact on communities, ecosystem and critical instruction in the event of catastrophic failure been undertaken and to reflect final conditions? If so, when did this assessment take place?	Yes, 2017	Yes, 2018	Yes, 2019	No
18	Is there (a) a closure plan in place for this dam (b) does it include long term monitoring	Yes	Yes	Yes	Yes
19	Have you, or do you plan to, assess your tailings facilities against the impact of more regular extreme weather events as a result of climate change, e.g. over the next two years?	Yes	Yes	Yes	Yes
20	Any other relevant information and supporting documentation.	Tailings volume reduced through consumption in underground mine paste fill	Tailings volume reduced through consumption in underground mine paste fill	Presently dry stack is being done for tailing storage. Also, started backfilling in mines through hydro fill and paste fill plant	Disposal Stopped since July 2019. External inspection carried out and capping is planned